

THE EFFECT OF DIFFERENT TRAINING PROGRAMS ON ECCENTRIC ENERGY UTILIZATION IN COLLEGE-AGED MALES

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ABSTRACT

Hawkins, SB, Doyle, TLA, and McGuigan, MR. The effect of different training programs on eccentric energy utilization in college-aged males. *J Strength Cond Res* 23(7): 1996–2002, 2009—The Eccentric Utilization Ratio (EUR), which is the ratio of countermovement jump (CMJ) to squat jump (SJ) performance measures, is a useful indicator of training status in elite athletes and their utilization of the stretch-shortening cycle. This investigation sought to determine if EUR was sensitive to different types of resistance training in untrained college-aged males. Twenty-nine college-aged males completed 8 weeks of training and were randomly allocated to 1 of 3 training programs: weight training ($n = 10$), plyometrics ($n = 10$), or weightlifting ($n = 9$). Testing occurred 3 times (pre, mid, post) with a CMJ and SJ conducted on a force plate integrated with a position transducer. Height, weight, and a 1RM (repetition maximum) squat also were measured. Weightlifting significantly ($p < 0.05$) helped subjects jump higher and produce more power than plyometrics for height and power for both CMJ and SJ results. This investigation indicated EUR did not significantly change, suggesting that this type of performance indicator may not be useful in a recreationally active population. Alternatively, an 8-week training program might not be a long enough time period to see changes in this group of participants. Results did indicate that high-velocity and high-force training programs, consisting of weightlifting and plyometrics, improved lower-body performance, especially in the areas of jump height and power.

KEY WORDS exercise test, muscle development, weightlifting

INTRODUCTION

The Eccentric Utilization Ratio (EUR), which is the ratio of a countermovement jump (CMJ) to squat jump (SJ) performance measures, has been shown to be useful as an indicator of training status in elite athletes (4,8). To date, EUR seems to be sensitive to different phases of resistance training programs (10). Generally, an athlete that has been training explosively (e.g., in their power phase of training) will have an EUR greater than 1 and an athlete performing traditional heavy load (e.g., in their hypertrophy phase) will have an EUR less than 1. Doyle's (4) investigation showed that there was an increase in the participant's EUR from pre training testing to post training testing, and this work was extended upon and supported by McGuigan et al. (8). McGuigan and associates (8) also concluded that the EUR was an indicator of the stretch-shortening cycle (SSC) performance. In a review of strength training and its role in vertical jump (VJ) performance, Baker (1) reported that presumably a strength training program that trains the efficiency of the SSC will improve VJ performance. In Baker's review he identified the different purposes of different styles of strength training. He noted that general strength training seemed to be inferior to general dynamic training in terms of improving VJ performance; although the inclusion of squats was important for VJ performance increases. These different types of strength, or resistance, training are further considered later in this article.

Weight-training studies report significant increases in muscular strength and upper-body power (10). Further investigations (2,3) suggest weight training can increase muscle size, power production, and strength of the lower body. Plyometrics, a general dynamic type of training, enables an athlete to improve power output and increase explosiveness. Results from Potteiger et al. (11) were similar to Luebbbers and associates (7) in that both studies showed an increase in VJ height and power. Weightlifting has been proposed by Newton and Kraemer (9) as an effective exercise for developing explosive power of the lower body. An athlete's lower-body strength and power, and VJ performance, may benefit from weightlifting as part of their strength

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23(7)/1996–2002

Journal of Strength and Conditioning Research
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training program because of the specificity of the loads and velocity of the movement. Garhammer (5) found similarities between the propulsive force patterns of weightlifting movements and the VJ. Results from these studies suggest that weight training, plyometrics, and weightlifting programs can have a positive effect on lower-body strength and power and VJ height and power. Furthermore, these results suggest that weightlifting and plyometrics have a greater potential to improve performance compared to weight training.

Doyle's (4) research in which 11 junior elite athletes were trained using a periodized program over a 34-week timeline studied the effects of training, on EUR, for jump height, power output, and power output relative to bodyweight. The results of the investigation showed that there was an increase in the participant's EUR from pre training testing to post training testing, with significant results found in the final testing phase. Doyle (4) concluded that the results presented in this study positively supported the goals of the training program, which were to increase strength and power output, as confirmed by an increase in the EUR for the tested variables across the participant group. They also support the suggestions of Baker (1), that a general strength training program was inferior for improving VJ compared to a more dynamic type of strength training.

McGuigan and colleagues (8) compared the EUR of athletes from different sports in different phases of training. They used elite athletes from rugby union, Australian rules football (ARF), soccer, softball, and field hockey. The variables measured were jump height and peak power. The results showed a significant difference from off season to preseason of EUR height for rugby union and EUR power for field hockey. McGuigan et al. (8) concluded that in sports such as rugby union, AFL, and soccer the athletes appear to have higher EUR values compared with other sports, which may correlate to the type of activity being heavily reliant on the athlete's ability to utilize the SSC. According to McGuigan et al. (8), the EUR provides the practitioner with information about the SSC performance of athletes and appears to be sensitive to changes in the type of training being undertaken. EUR can be used to determine the effectiveness of resistance training on lower-body strength and power. By monitoring the EUR, a coach or sport scientist may be able to track the changes in lower-body strength and power of participants undertaking different training programs.

Therefore, the primary purpose of this investigation was to identify if EUR was sensitive to the different types of resistance training programs (weight training, plyometrics, and weightlifting). A secondary purpose was to identify if the different training programs have influences on VJ performance and lower-body strength. It is hypothesized that EUR will be higher in the dynamic strength training groups compared to the general strength training group. Similarly, it is also hypothesized that CMJ and VJ performance will be

superior in the dynamic strength training groups compared to the general strength training groups.

METHODS

Experimental Approach to the Problem

This investigation compared 3 different training programs—weight training (a general strength training program), plyometrics, and weightlifting (both dynamic strength training programs)—and the effects that these resistance training programs had on the EUR and VJ performance and lower-body strength. Testing was completed at week zero (pre), week 5 (mid), and week 9 (post) to monitor changes over the course of an 8-week training program. A CMJ and SJ were completed on a force plate interfaced with a position transducer to determine displacement and vertical ground reaction forces. Jump height and power were determined, and a standard VJ, together with the 1RM (repetition maximum) squat strength, was measured. This would allow coaches and other professionals to have a simple measure that could be used to monitor the training progress of athletes in their charge.

Subjects

Twenty-nine nonathlete college-aged males (age 21.5 ± 12.5 years, height 1.80 ± 0.21 m, mass 77.5 ± 15.7 kg) were recruited and randomly assigned into 1 of 3 training groups: weight training ($n = 10$), plyometrics ($n = 10$), and weightlifting ($n = 9$). Each participant was required to sign an informed consent letter prior to commencement of the investigation. The inclusion criteria stated that participants must not be currently involved in any form of organized, competitive training or sport. Participants also must not have undertaken any type of weight training within the last 6 months. This research was given clearance from the university human research ethics committee.

Procedures

Training. Participants completed 2 familiarization sessions; both sessions introduced participants to the technically difficult exercises. The second familiarization session also was used to perform the initial testing. Participants were required to attend 3 training sessions per week; each session lasted about an hour. During week 5 of training, the mid training testing was conducted; post training testing occurred in week 9. All training programs were periodized and consisted of two 4-week macrocycles increasing in difficulty and intensity (Tables 1–3). The training programs (weight training, plyometrics, and weightlifting) for this study were designed to represent typical training programs of each type of resistance training; therefore, the training volumes have not been equated between the different training regimens because the training volume was not the basis of the study, but it was the type of training that was being investigated. Load was increased when participants were easily able to complete all repetitions for each set of the exercise. The participants' starting load was determined during the first familiarization session.

TABLE 1. Traditional weight training program.

Weeks	Day 1		Day 2		Day 3
	Exercise	Reps	Exercise	Reps	
1-4	Squats	8 8 10	Deadlifts	8 8 10	As per Day 1
	Lunges	6 6 6	Stiff-leg deadlifts	8 6 6	
	Bent-over row	8 8 10	Step-ups (dumbbell)	8 8 8	
	Pullover	8 8 8	Walking lunges (dumbbell)	6 6 6	
	Upright row	8 8 8	Calf raises (Stand.)	10 10 10	
5-8	Squats	8 6 4	Split squats	8 8 6	As per Day 1
	Lunges	6 6 6	Stiff-leg deadlifts	8 6 6	
	Prone pull-ups	8 6 6	45-degree lunges	6 6 6	
	Bench press	8 6 4	Standing 1-arm row	8 6 6	
	Good mornings	6 6 6	Standing dumbbell press	8 6 6	

Testing. Participants were asked to perform a CMJ and an SJ with a lightweight bar across their shoulders. Each jump was completed on a 400 Series Performance Force Plate (Fitness Technology, Skye, South Australia), which recorded vertical ground reaction forces, with the lightweight bar attached to a position transducer (model PT9510; Celesco, Canoga Park, California, USA) to measure jump displacement. Both the force plate and position transducer were interfaced with the Ballistic Measurement System (BMS; Fitness Technology). Reliability of this system has been reported previously by McGuigan and associates (8), who reported an intraclass correlation coefficient of $R > 0.96$ and coefficient of variation of $< 3\%$.

Participants were instructed to complete a standard CMJ in which they were to squat down into a self-selected depth prior to explosively performing the concentric action of the CMJ. The SJ was completed by the participant holding a self-selected squat position for 4 seconds before completing a concentric-only SJ. For both jumps the participant was

instructed to jump as high as possible. VJ was recorded using a Swift Yardstick (Lismore NSW, Australia). Participants were allowed to use an arm swing for this test. The best of 3 trials was recorded. A 1RM squat (with the participant's thighs being parallel to the floor, to complete a successful lift) and participant's height and mass were recorded at each testing phase.

Analysis

BMS software provided displacement and vertical ground reaction force data, and from these measurements velocity, force, and power were analyzed using custom-written Matlab software (The Mathworks, Natick, Massachusetts, USA).

$$v_{(t)} = dx/dt$$

$$P_{(t)} = F_{(t)} * v_{(t)}$$

where x = displacement, v = velocity, t = $1/\text{sampling frequency}$, F = force, and P = power.

TABLE 2. Plyometrics training program.

Weeks	Day 1		Day 2	Day 3
	Exercise	Reps		
1-4	Jumps in place	8 8 8	As per Day 1	As per Day 1
	Squat jumps	6 6 6		
	A-skips	10 10 10		
	Single-leg hops	6 6 6		
	Double-leg hops	3 3 3		
5-8	Double-leg hops	5 5 5	As per Day 1	As per Day 1
	Jump and reach	6 6 6		
	Double-leg tuck jump	6 6 6		
	Depth jump	5 5 5		
	Bounding	15 15 15		

For bounding, participants bounded over 15 meters, not 15 reps.

TABLE 3. Weightlifting training program.

Weeks	Day 1		Day 2	Day 3	
	Exercise	Reps		Exercise	Reps
1–4	High pull	6 6 6	As per Day 1	High pull	6 6 6
	Hang clean	5 5 5		Hang clean (above knee)	5 5 5
	Power push (front)	5 5 5		Push press (front)	5 5 5
	Snatch balance	5 5 5		Jump squats	5 5 5
	Front squats	8 6 6		Back squats	8 6 6
5–8	Hang clean (below knee)	5 5 3	As per Day 1	Hang clean (below knee)	5 5 3
	Split jerks	5 5 3		Split jerks	4 4 4
	Push press (front)	5 5 5		Snatch balance	4 4 4
	Jump squats	5 5 3		Speed squats	5 5 5
	Front squats	6 4 2		Back squats	6 4 2

Power was determined during the propulsion phase of the SJ and CMJ. The EUR for height and power was calculated by averaging the maximum value from all of the 3 trials; this also was completed using custom-written Matlab software. McGuigan et al (8) derived EUR from height and power; consequently, EUR was calculated from these same 2 variables in this investigation.

Statistical Analyses

Two-way repeated measures analysis of variance (ANOVA) (training program $\times$ time) were performed to identify significant differences between and within groups. An alpha level of 0.05 was used as the criterion to determine significance. Post hoc tests were used to determine where the differences existed. Effect size was calculated to indicate the magnitude of difference (11).

RESULTS

Eccentric Utilization Ratio

The EUR results showed no significant differences in height and power performance variables ($p > 0.05$) at any testing phases. Table 4 shows the full details of EUR results. The results did show a nonsignificant increase in height and power for the weight-training group from pre to mid testing; also the plyometric group had a nonsignificant increase in height EUR mid to post testing. The weightlifting group had a nonsignificant overall increase of EUR for power from pre to post testing. In post testing effect sizes ranged from 0.7 to 1.3 among the 3 training groups, indicating a very large effect size (6).

Countermovement Jump

A main effect of the training group for CMJ results for height ($p < 0.05$) and power ($p < 0.05$) was seen; these differences lay between the plyometrics and weightlifting group. At the pre training phase there was a significant difference ($p < 0.05$) between the plyometric group and the weightlifting group for

height, which also had a moderate effect size (0.74). At the mid training testing phases, there was a significant difference and a large effect size between the weight-training and plyometrics groups for the height results ($p < 0.05$). The detailed results for CMJ are shown in Table 5. The weightlifting group, however, showed a significant increase from pre training to post training in power ($p < 0.05$); this had a moderate effect size (0.66). From pre to mid testing for the weight-training group, there was a significant ($p < 0.05$) increase in height with a large effect size (1.41).

Squat Jump

The SJ results are shown in Table 6. Statistical analysis revealed significant differences for power ($p < 0.05$) between the weight-training and plyometric groups, which had a large effect size (1.55) at the pre testing phase. The plyometric and

TABLE 4. EUR results, as measured for height and power, mean ($\pm$ SD). There were no significant differences.

	Group	n	Height (AU)	Power (AU)
Pre testing	WT	10	1.20 $\pm$ 0.09	1.01 $\pm$ 0.09
	Plyo	10	1.23 $\pm$ 0.20	1.08 $\pm$ 0.15
	Lift	9	1.09 $\pm$ 0.23	0.99 $\pm$ 0.09
Mid testing	WT	10	1.25 $\pm$ 0.16	1.06 $\pm$ 0.13
	Plyo	10	1.18 $\pm$ 0.25	1.05 $\pm$ 0.12
	Lift	9	1.06 $\pm$ 0.05	1.06 $\pm$ 0.15
Post testing	WT	10	1.12 $\pm$ 0.10	0.98 $\pm$ 0.06
	Plyo	10	1.22 $\pm$ 0.18	1.02 $\pm$ 0.08
	Lift	9	1.05 $\pm$ 0.06	1.00 $\pm$ 0.07

WT, weight training; Plyo, plyometrics; lift, weightlifting.

TABLE 5. Countermovement jump results for height and power (mean $\pm$ SD).

	Group	<i>n</i>	Height (m)	Power (W)
Pre testing	WT	10	0.42 $\pm$ 0.03	5360 $\pm$ 272 [†]
	Plyo	10	0.38 $\pm$ 0.06*	4564 $\pm$ 538 [†]
	Lift	9	0.44 $\pm$ 0.10*	5653 $\pm$ 1028 [†]
Mid testing	WT	10	0.47 $\pm$ 0.04* [†]	5614 $\pm$ 962
	Plyo	10	0.38 $\pm$ 0.06* [†]	4969 $\pm$ 913
	Lift	9	0.44 $\pm$ 0.05	5836 $\pm$ 846
Post testing	WT	10	0.42 $\pm$ 0.11	5128 $\pm$ 895 [†]
	Plyo	10	0.40 $\pm$ 0.09	4592 $\pm$ 909 [†]
	Lift	9	0.46 $\pm$ 0.04	6336 $\pm$ 1028 [†]

WT, weight training; Plyo, plyometrics; lift, weightlifting.

*A significant ($p < 0.05$) interaction of time and training group.[†]A significant ($p < 0.05$) main effect of training group.**TABLE 6.** Squat jump results for height and power (mean $\pm$ SD).

	Group	<i>n</i>	Height (m)	Power (W)
Pre testing	WT	10	0.35 $\pm$ 0.03	5338 $\pm$ 526*
	Plyo	10	0.31 $\pm$ 0.06	4293 $\pm$ 793*
	Lift	9	0.41 $\pm$ 0.07	5777 $\pm$ 1,223*
Mid testing	WT	10	0.38 $\pm$ 0.06	5286 $\pm$ 479
	Plyo	10	0.33 $\pm$ 0.07	4820 $\pm$ 1,108
	Lift	9	0.41 $\pm$ 0.04	5547 $\pm$ 843
Post testing	WT	10	0.37 $\pm$ 0.09	5236 $\pm$ 837
	Plyo	10	0.34 $\pm$ 0.09	4559 $\pm$ 1,052*
	Lift	9	0.44 $\pm$ 0.04	6343 $\pm$ 1,152*

WT, weight training; Plyo, plyometrics; lift, weightlifting.

*A significant ($p < 0.05$) interaction of time and training group.**TABLE 7.** 1RM squat and vertical jump results (mean $\pm$ SD).

	Group	<i>n</i>	1RM Squat (kg)	VJ (m)
Pre testing	WT	10	112.25 $\pm$ 13.25*	56.2 $\pm$ 2.9
	Plyo	10	86.50 $\pm$ 13.55*	51.9 $\pm$ 6.0
	Lift	9	116.25 $\pm$ 16.00*	57.9 $\pm$ 6.9
Mid testing	WT	10	130.50 $\pm$ 18.50*	57.3 $\pm$ 2.5
	Plyo	10	102.00 $\pm$ 16.50*	54.6 $\pm$ 6.8
	Lift	9	124.50 $\pm$ 19.00*	59.6 $\pm$ 6.6
Post testing	WT	10	147.75 $\pm$ 20.50*	59.4 $\pm$ 4.6
	Plyo	10	111.00 $\pm$ 13.50*	55.5 $\pm$ 7.6
	Lift	9	142.50 $\pm$ 20.00*	62.8 $\pm$ 6.4

WT, weight training; Plyo, plyometrics; lift, weightlifting; VJ, vertical jump.

*Significantly different ($p < 0.05$) interaction of time and training group.

weightlifting group were significantly different ($p < 0.05$), with a large effect size (1.46) at the pre testing phase. Also, there was a significant difference ($p < 0.05$) and a large effect size (1.62) between the plyometric and weightlifting group at the post testing phase for power. The weight-training group had a nonsignificant increase in height from pre training to mid training, which had a moderate effect size (0.62). The weightlifting group saw no significant differences in the increases in values from mid training to post training for height with a moderate effect size (0.75). The weightlifting group had a nonsignificant increase, which was of a moderate magnitude of difference (0.79) from mid training to post training for power.

VJ and 1RM

At each of the testing periods 1RM results were significantly different between the plyometrics group and both the weight-training and weightlifting groups. Although not significant ($p = 0.06$), changes in VJ improved for each group at each time period, with the weightlifting group showing the greatest improvement; the weight-training group showed the least amount of improvement. Results are provided in Table 7.

DISCUSSION

The primary purpose of this investigation was to determine if the measure of EUR was sensitive to the different types of resistance training programs—weight training, plyometrics, and weightlifting—in untrained college-aged males. A secondary purpose was to identify if the different training programs have a different impact on VJ performance and lower-body strength. This investigation was the first to use a longitudinal design using recreationally active athletes to investigate the EUR. Doyle (4) and McGuigan et al. (8) conducted the research on professional and elite athlete groups; their results suggest that EUR is sensitive to changes in elite athletes.

Because EUR has been previously shown to be associated with increases in performance (4,8), it was hypothesized that EUR would be sensitive to the different types of resistance training. The weightlifting group had a nonsignificant increase in the EUR for height and power compared to the other 2 training groups. The trends indicate, however, that this type of high-velocity, high-force power training had an effect on lower-body performance. This lack of significance is possibly a result of the limited number of participants and duration of the program. The other 2 resistance training programs also showed similar trends for the EUR results.

Based on the current results, EUR does not seem to be sensitive in an untrained athlete population over an 8-week training period. Thus, the first hypothesis was not supported. However, some trends indicated a change in EUR for both height and power. These trends may have proved significant with more participant numbers or a training period longer than 8 weeks. Alternatively, it may indicate poor coping with the training for all the athletes because of the lack of physical preparation and their current level of physical conditioning.

These results suggest that in this population, changes may take a longer time to be realized, so an 8-week training period might be insufficient to see changes in the EUR measure in this population. Further, the differences between the groups at baseline may have also played a role in the lack of significant results. Further investigations are necessary in this area of EUR to determine its sensitivity in a similar population of athletes.

Typically before undertaking a plyometrics and weightlifting program, athletes would first partake in a weight-training program to help prepare them for the type of training involved in plyometrics and weightlifting. Certain strength criteria, such as ability to squat a certain weight or bench press a certain weight, must be met before an athlete undertakes a plyometric or weightlifting type of resistance training. In this instance, however, this did not occur, so this might also be partially responsible for the lack of significant findings for the EUR because both the plyometric and weightlifting groups did not complete a strength building phase prior to their programs. However, it should be noted that both the CMJ and SJ measures from the pre to post training phases did match what would be expected from both the plyometrics and weightlifting groups, but the results did not reach statistical significance for EUR. It also should be noted that the programs were conservatively progressive in overloading the participants, and this also may have played a role in the lack of significant findings.

Overall the trends indicate that weightlifting had a greater effect on lower-body performance across both the CMJ (Table 5) and the SJ (Table 6) compared to the weight-training group and the plyometrics group. Similar nonsignificant trends were evidenced for the VJ results. These results suggest that the high-velocity and high-force type of training typically used in a weightlifting training program has a more pronounced effect on physical performance of the lower-body power measures than do the other 2 groups. This finding supports the second hypothesis that a dynamic strength training program will be more beneficial to CMJ and VJ performance. Because the results did not reach statistical significance for EUR in this investigation, which was in contrast to earlier research, this suggests that the use of EUR for professional or semi-professional athletes is more sensitive to changes than for an untrained population; alternatively, a longer training period may be required for this group. The findings from the 1RM results are interesting in that all groups improved their strength as measured by a 1RM squat, including the plyometric group, who did not train with any weights. It also is encouraging that all 3 types of training improved VJ performance, although arguably the weight-training group was less effective than the other 2 groups, in particular the weightlifting group.

PRACTICAL APPLICATIONS

Tracking an athlete's progress throughout a training program is an important process for strength and conditioning

specialists and sport scientists. The EUR can be used to provide practitioners with information regarding an SSC performance, suggested by Doyle (4). When undertaking a training program to increase lower-body strength and power in an athlete, this investigation suggests that weightlifting and plyometrics seem to be effective in improving the lower-body strength and power performance of recreationally active athletes. This type of combination training, including high-velocity and high-force exercise, can lead to the increase in power performance of the athlete. Although the EUR was not significant in this study, it is suggested that over a longer training period, it might provide a means by which to monitor progress of the novice athletes as can currently be done in elite athletes. This study, along with the investigations by Doyle (4) and McGuigan et al. (8), suggests that the use of EUR is for monitoring an athlete's progress with a training program. Therefore, EUR does not necessarily have to increase because different training goals will result in varying lower limb performance. As with any performance measure used for monitoring an athlete, EUR needs to be viewed as part of an overall picture of an athlete profile and not just in isolation. The findings from this study also support previous suggestions that a dynamic type of strength training is more beneficial to improving jumping performance than a general strength training program (1).

ACKNOWLEDGMENTS

The authors thank Mr. David Kinsella for his help with data collection and assistance with training the participants.

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